Signal Propagation in Single Cycle RV32I

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1. Introduction

This report documents the comprehensive testing and validation of a 32-bit RISC-V processor implemented in CircuitVerse. The objective is to verify correct operation of all major processor components and confirm proper instruction execution across the entire processor pipeline.

The entire evolution of this document and project is available in our GitHub repository.

The project is organized into three phases:

1. **Part 1 - Instruction Memory Decoding**: Identification and analysis of the instruction set stored in the processor's instruction memory
2. **Part 2 - Individual Component Testing**: Systematic verification of each processor subsystem in isolation
3. **Part 3 - Processor Integration Testing**: End-to-end validation of the complete processor executing the full instruction sequence

Each phase includes detailed test protocols, measurements at critical signal points using the CircuitVerse Flag element, and verification that each component or instruction meets its functional specification.

The processor operates on 8-bit data but uses 32-bit instruction and program counter (PC) values, as specified in the reference design. Flags are strategically placed throughout the design to enable real-time signal observation and verification during simulation.

2. Processor Architecture Overview

The RISC-V processor implemented in this project follows a modular design with distinct functional blocks that interact through well-defined signal interfaces. Understanding the architecture is essential for developing effective test strategies and interpreting component behavior in context.

2.1. System Architecture

The processor is composed of the following primary subsystems:

- **Instruction Memory (instr_mem)**: Holds the program instructions to be executed. Stores 8 words of 32-bit instruction data.

- **Register File (reg_file)**: Provides 8 registers for operand storage and intermediate computation results.
- **Control Unit (control_unit)**: Decodes instructions and generates control signals that coordinate subsystem operation.
- **Immediate Generator (imm_gen)**: Extracts and extends immediate values embedded within instructions.
- **Arithmetic Logic Unit (ALU)**: Performs arithmetic and logical operations on operands.
- **Data Memory (data_mem)**: Stores and retrieves program data during execution.

These components operate synchronously, with state changes occurring on clock edges. The processor employs synchronous reset, requiring the reset signal to be asserted high followed by a clock pulse to reinitialize the PC and memory elements.

2.2. Data Path and Signal Flow

Instructions are fetched sequentially from instruction memory based on the current PC value. The control unit decodes each instruction, extracting opcode, function codes, and register indices. Operands are retrieved from the register file, immediate values are generated as needed, and the ALU or data memory perform the specified operation. Results are written back to the register file, and the PC advances to the next instruction address.

Throughout execution, key signal points such as instruction values, operand data, ALU results, and memory access addresses are available for measurement and verification, enabling detailed observation of processor operation at any instruction cycle.

3. Part 1: Instruction Memory Decoding

The instruction memory block contains the program instructions that define the processor's behavior during validation.

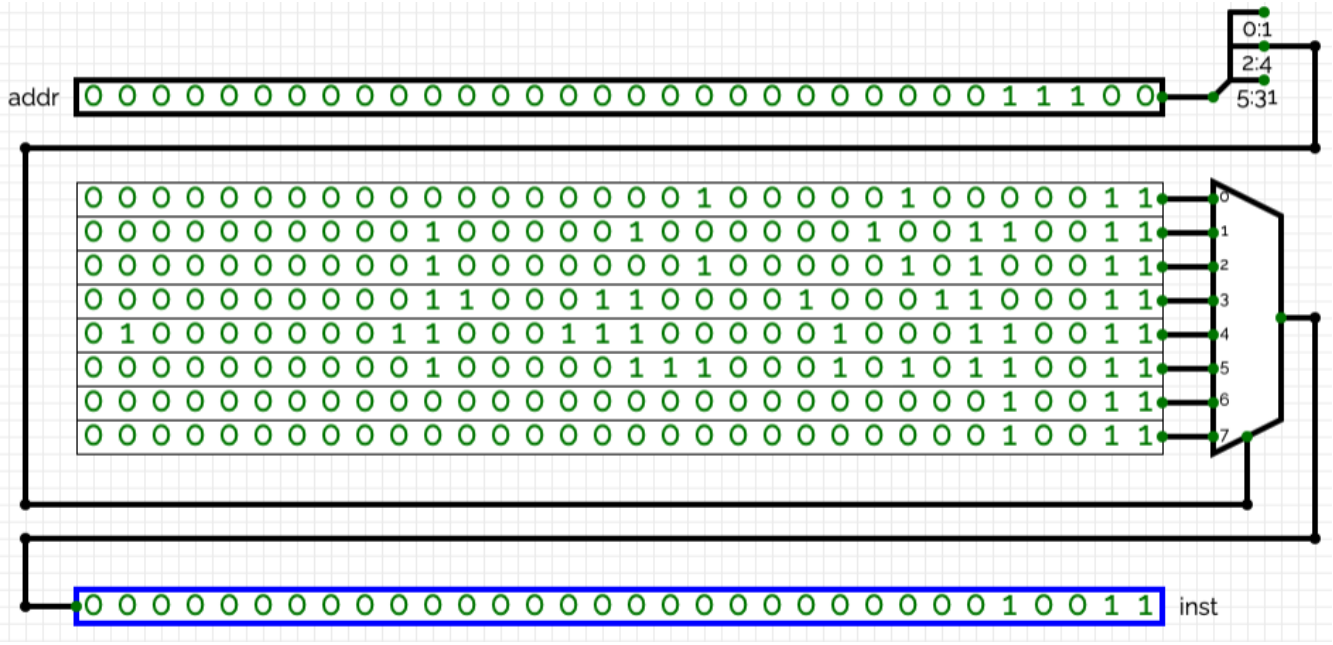


Figure 1. Instruction Memory Circuit Diagram

3.1. Decoding Protocol

To decode the instruction memory contents, we accessed the `instr_mem` block within the CircuitVerse design and observed the binary instruction values stored at each memory address. Each 32-bit instruction follows the RISC-V instruction encoding format, with specific bit fields defining the operation, source registers, destination register, and immediate values.

The decoding process involved:

1. Identifying the opcode (bits [6:0]) to determine instruction type
2. Extracting the rd (destination register), rs1 (source 1), and rs2 (source 2) fields
3. Parsing function code fields (funct3, funct7) that refine the operation
4. Decoding immediate values for I-type, S-type, and B-type instructions
5. Translating the binary encoding to RISC-V assembly code

3.2. Documented Instructions

The following table shows the decoded instruction memory contents. The instruction memory stores eight 32-bit instructions, each located at word-aligned addresses from `0x00` to `0x1C`.

Table 1. Instructions

PC	Binary Instruction	Hex	Format	Assembly Instruction
`0x00`	`00000000000000000010000010000011`	`0x00002083`	I-type	`lw x1, 0(x0)`
`0x04`	`00000000001000000100000100110011`	`0x00208133`	R-type	`add x2, x1, x2`
`0x08`	`00000000001000000001000010100011`	`0x002020A3`	S-type	`sw x2, 1(x0)`

PC	Binary Instruction	Hex	Format	Assembly Instruction
0x0C	00000000001100011000010001100011	0x00318463	B-type	beq x3, x3, 8
0x10	01000000011000111000001000110011	0x40638233	R-type	sub x4, x7, x6
0x14	00000000001000001110001010110011	0x0020E2B3	R-type	or x5, x1, x2
0x18	00000000000000000000000000010011	0x00000013	I-type	addi x0, x0, 0 (nop)
0x1C	00000000000000000000000000010011	0x00000013	I-type	addi x0, x0, 0 (nop)

The instruction sequence executed by the processor is:

```
0x00: lw   x1, 0(x0)
0x04: add  x2, x1, x2
0x08: sw   x2, 1(x0)
0x0C: beq  x3, x3, 8
0x10: sub  x4, x7, x6
0x14: or   x5, x1, x2
0x18: addi x0, x0, 0
0x1C: addi x0, x0, 0
```

The branch instruction at address `0x0C` compares `x3` with itself, so the branch condition is true. With an immediate value of `8`, the branch target is `0x14`, meaning that the instruction at `0x10` is skipped when the branch is taken.

The final two instructions are `addi x0, x0, 0`, which is the RISC-V no-operation instruction (`nop`). Since `x0` is always zero, these instructions do not modify the processor state.

The identified instructions form the test sequence executed during Part 3 processor validation.

4. Part 2: Individual Component Testing

Component-level testing validated that each processor subsystem functions correctly in isolation before integration into the complete data path. This phase ensures that issues are identified and resolved at the component level, simplifying debugging of whole-processor behavior.

The testing methodology for each component involved:

1. Developing a comprehensive test suite using the CircuitVerse TESTBENCH that exercises various operational modes and edge cases
2. Observing test results and ensuring they all pass
3. Documenting results to provide evidence of correct operation
4. Exporting test cases as csv files to save for future use and documentation

NOTE	Some input bits are marked with "X", meaning they are treated as don’t-care bits, this implies that injecting either "0" or "1" at the corresponding position yielded the exact same result for that test case.

The following subsections document the testing of each major component:

4.1. Instruction Memory

The Instruction Memory is a read-only combinational memory that provides a 32-bit instruction word for a given 32-bit address. It is used by the instruction fetch stage to supply the next instruction based on the program counter.

4.1.1. Specifications

Below we see the Instruction Memory circuit in CircuitVerse, with the input and output signals labeled for clarity:

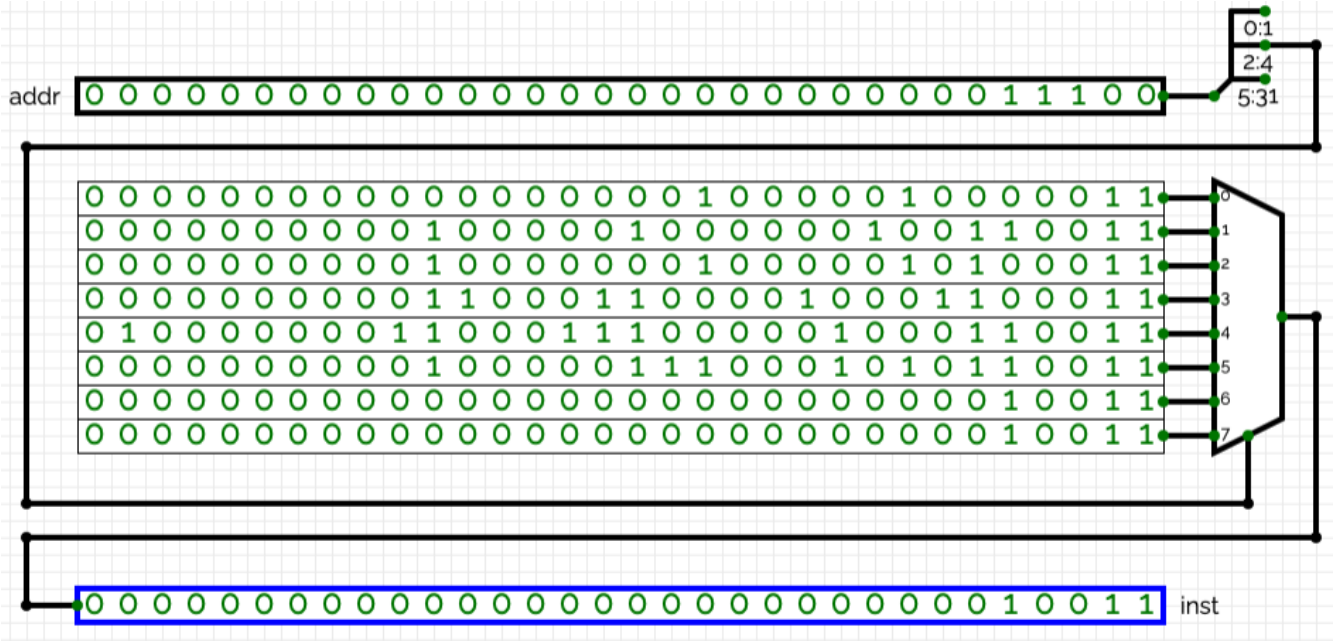


Figure 2. Instruction Memory Circuit Diagram

- **addr** (32-bit): Byte address input (word-aligned expected).
- **inst** (32-bit): Instruction word output stored at the given address.

The component performs a direct address-to-word lookup, addresses are expected to be aligned to 4-byte boundaries.

4.1.2. Testing Methodology

Verification supplies known 32-bit addresses and checks that the component returns the exact 32-bit instruction stored at each location. Tests exercise a contiguous region of addresses at the start of memory to confirm correct storage and address decoding.

4.1.3. Test Cases and Results

Table 2. Instruction Memory Test Cases

Input addr	Output inst
00000000000000000000000000000000 (0)	00000000000000000010000010000011
00000000000000000000000000000100 (4)	00000000001000001000000100110011
00000000000000000000000000001000 (8)	00000000001000000010000010100011
00000000000000000000000000001100 (12)	00000000001100011000010001100011
00000000000000000000000000010000 (16)	01000000011000111000001000110011
00000000000000000000000000010100 (20)	00000000001000001110001010110011
00000000000000000000000000011000 (24)	00000000000000000000000000010011
00000000000000000000000000011100 (28)	00000000000000000000000000010011

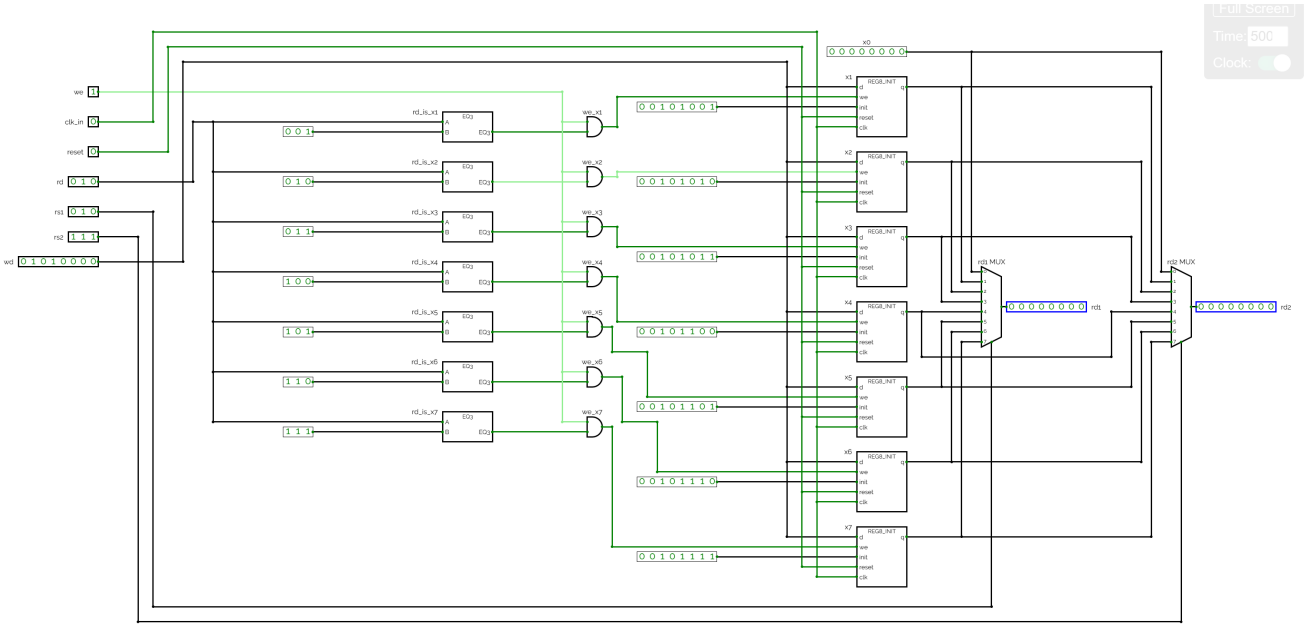
4.1.4. Verification

All supplied addresses returned the expected 32-bit instruction words. The Instruction Memory lookup and address decoding are verified for the tested address range.

4.2. Register File

The Register File is a sequential component used to store and provide access to the processor registers. In this simplified RV32I single-cycle processor, the register file contains eight 8-bit registers. The component has two read ports and one write port, allowing the processor to read two source operands and write one destination value.

4.2.1. Specifications



The Register File component has the following input and output signals:

Table 3. Register File Signals

Signal	Bitwidth	Description
`we`	1	Write enable signal. When `we = 1`, the selected destination register is updated on a clock pulse.
`clk_in`	1	Clock input. Writes and reset behavior occur synchronously when the clock is triggered.
`reset`	1	Synchronous reset signal. When enabled and the clock is triggered, the register file returns to its initialized values.
`rd`	3	Destination register address used for writing.
`rs1`	3	Source register address for the first read port.
`rs2`	3	Source register address for the second read port.
`wd`	8	Write data input. This is the value stored into the selected register when `we = 1`.
`rd1`	8	First read data output, selected by `rs1`.
`rd2`	8	Second read data output, selected by `rs2`.

The read ports are combinational, meaning that `rd1` and `rd2` change when `rs1` and `rs2` are changed. The write operation is synchronous, so the selected register only changes after a clock pulse when `we = 1`.

4.2.2. Testing Methodology

The Register File was tested inside CircuitVerse by manually changing the input values and observing the outputs `rd1` and `rd2`.

The tests were designed to verify the following behavior:

1. The register file responds correctly to reset.
2. The read ports correctly output the values selected by `rs1` and `rs2`.
3. A register can be written when `we = 1` and the clock is triggered.
4. A register is not modified when `we = 0`, even if a clock pulse is applied.
5. The register file can return to its initialized state after reset.

The complete test cases are documented in `tests/Register File Tests.csv`.

4.2.3. Test Cases and Results

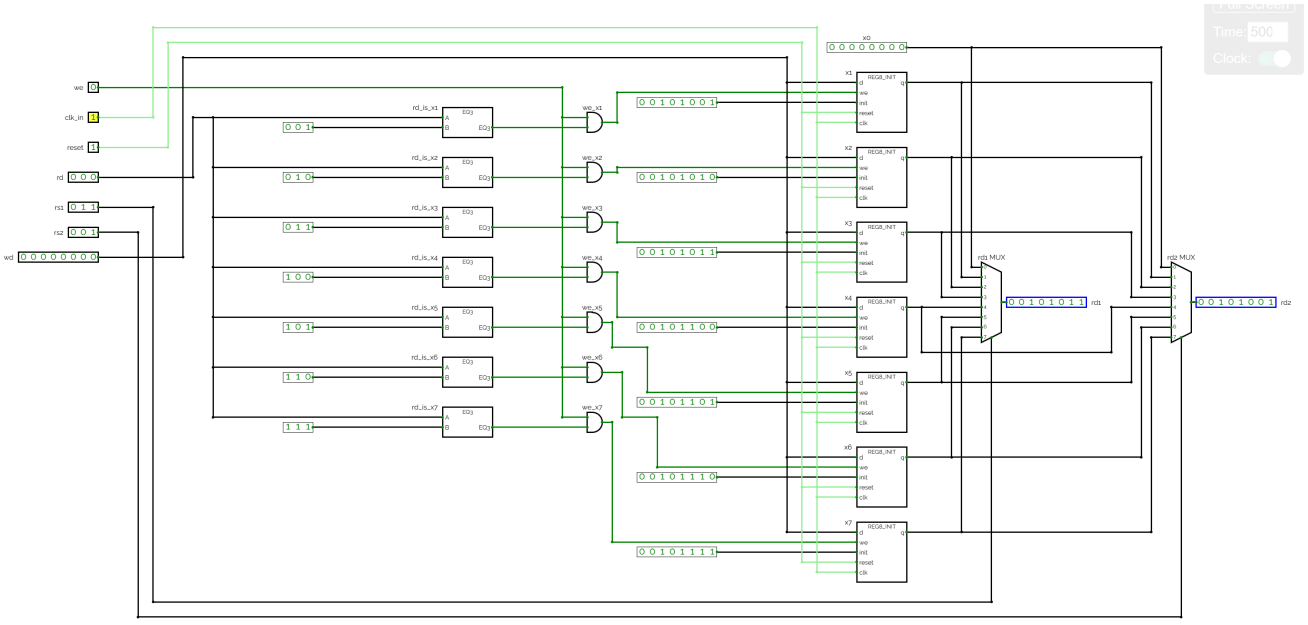
Table 4. Register File Test Cases

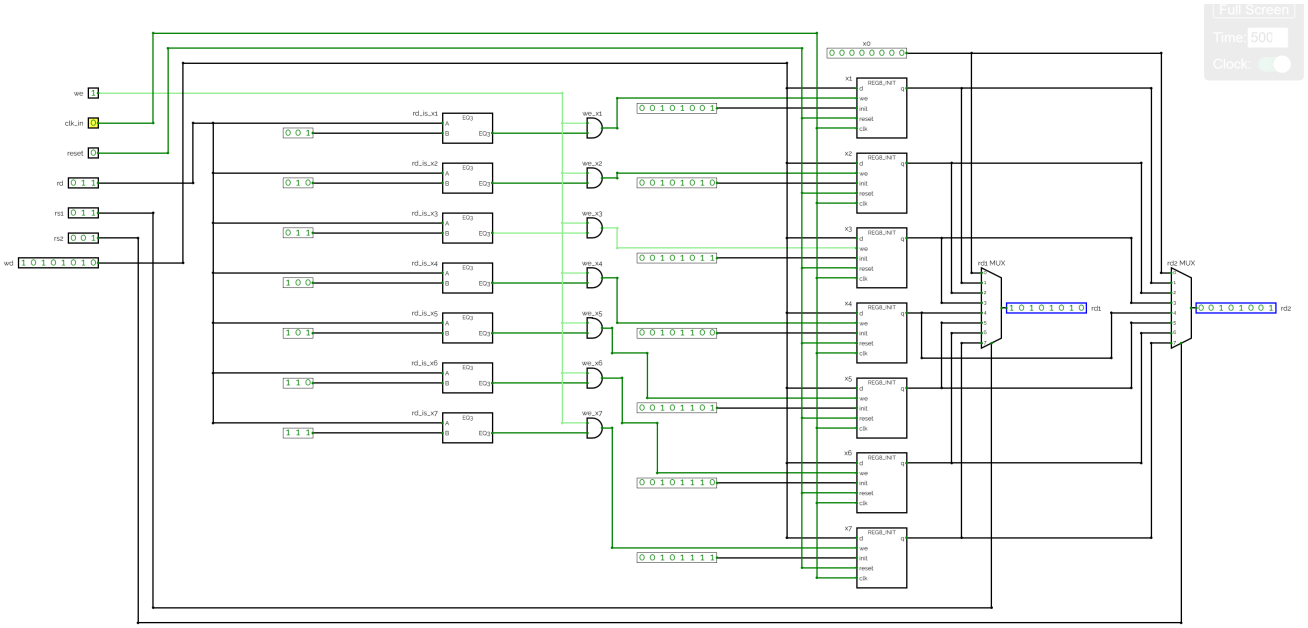
Test	Inputs Applied	Expected Output	Result
RF_RESET_X3_X1	`we=0`, `reset=1`, `rd=000`, `rs1=011`, `rs2=001`, `wd=00000000`, clock pulse	`rd1=00101011`, `rd2=00101001`	Passed

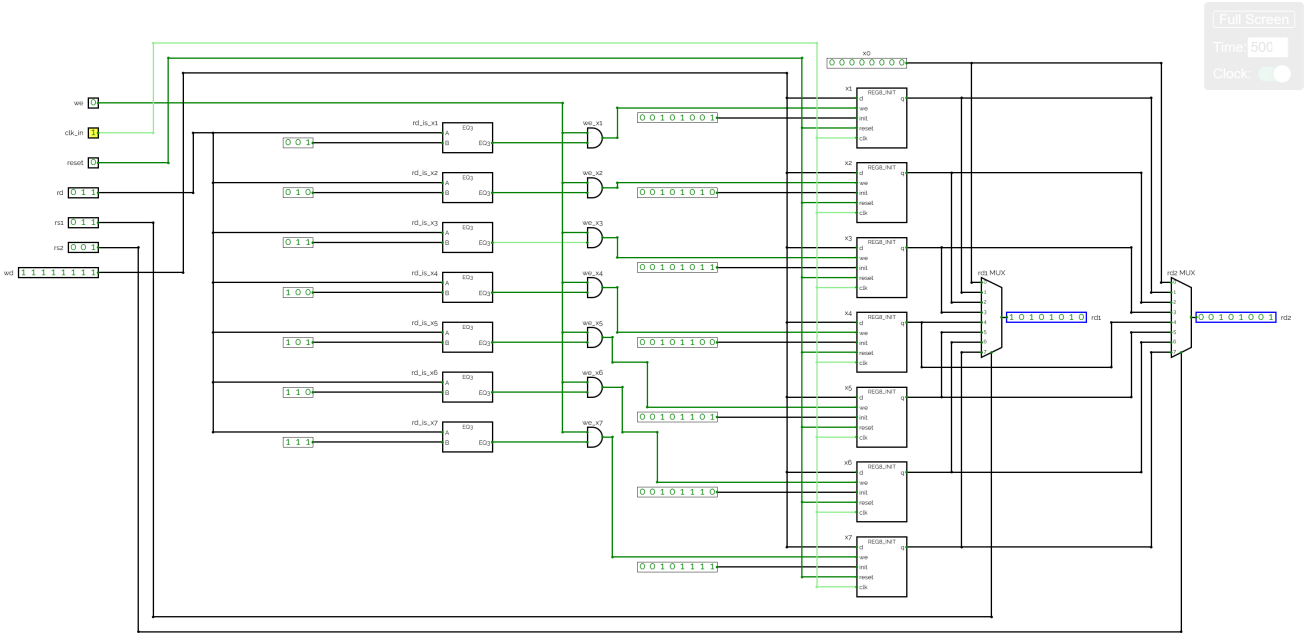
Test	Inputs Applied	Expected Output	Result
RF_WRITE_X3_AA	we=1, reset=0, rd=011, rs1=011, rs2=001, wd=10101010, clock pulse	rd1=10101010, rd2=00101001	Passed
RF_BLOCK_WRITE_X3	we=0, reset=0, rd=011, rs1=011, rs2=001, wd=11111111, clock pulse	rd1=10101010, rd2=00101001	Passed
RF_RESET_AFTER_WRITES	we=0, reset=1, rd=000, rs1=011, rs2=001, wd=00000000, clock pulse	rd1=00101011, rd2=00101001	Passed

4.2.4. Evidence

The following images show the internal CircuitVerse tests used to validate the Register File behavior.







4.2.5. Verification

All test cases produced the expected register file outputs. The reset test confirmed that the register file returns to its initialized values when `reset` is asserted and a clock pulse is applied. In this implementation, reading `x3` and `x1` after reset produces `rd1=00101011` and `rd2=00101001`.

The write test verified that register `x3` is updated to `10101010` only when `we = 1` and the clock is triggered. The disabled-write test confirmed that when `we = 0`, the register value remains unchanged even if a clock pulse is applied. Finally, the reset-after-write test confirmed that the register file can restore its initialized values after a previous write. Therefore, the Register File component is validated and ready for integration into the processor data path.

4.3. Control Unit

The Control Unit decodes the instruction opcode and function fields to generate the processor control signals. It translates `op`, `f3`, and `f7` inputs into branch, memory, ALU, and register control outputs used by the datapath.

4.3.1. Specifications

Below we see the Control Unit circuit in CircuitVerse, with the input and output signals labeled for clarity:

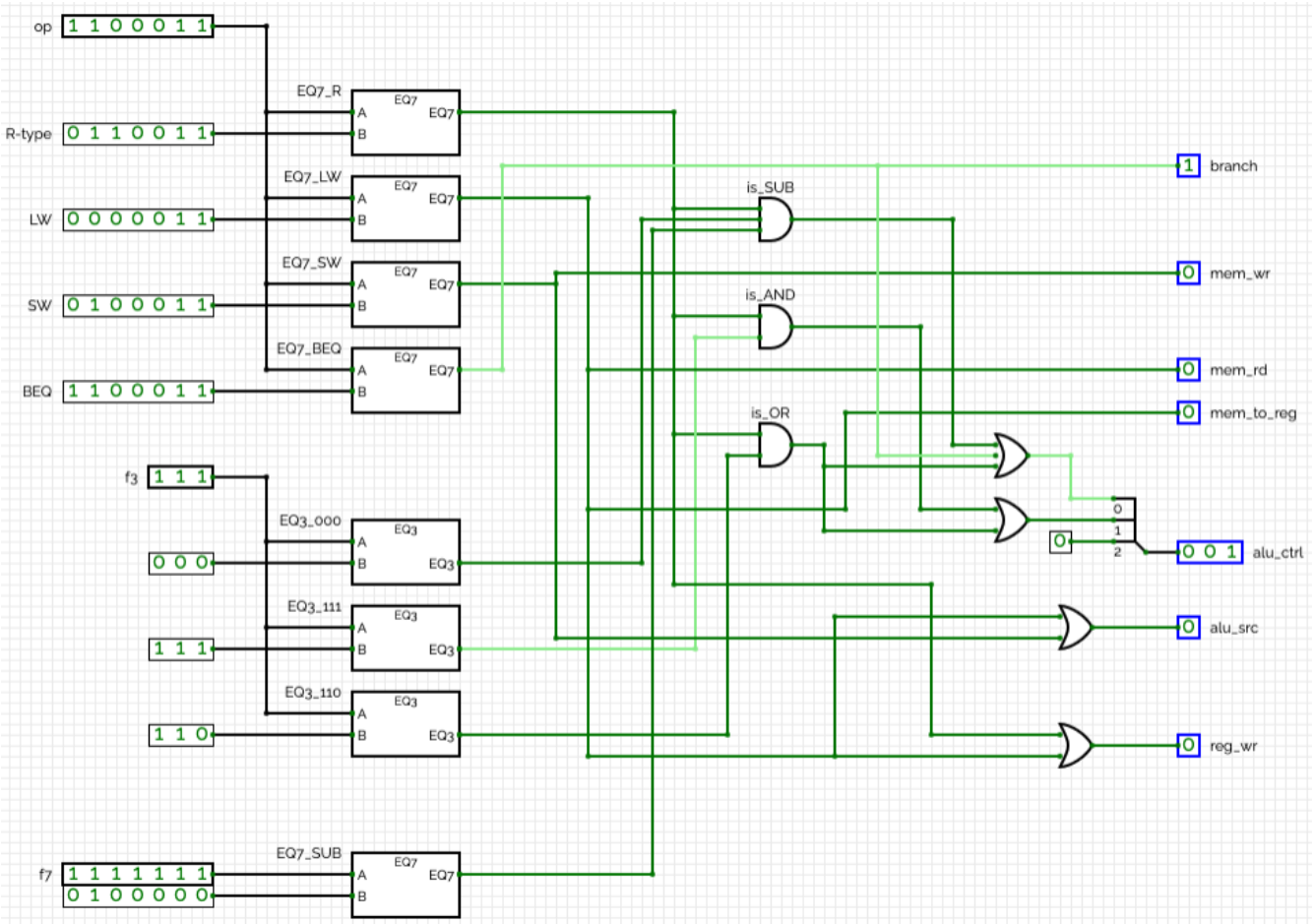


Figure 3. Control Unit Circuit Diagram

- **op** (7-bit): Opcode field from the instruction.
- **f3** (3-bit): funct3 field (function sub-code).
- **f7** (7-bit): funct7 field (function sub-code).
- **branch** (1-bit): Asserted for branch instructions.
- **mem_wr** (1-bit): Memory write enable for store instructions.
- **mem_rd** (1-bit): Memory read enable for load instructions.
- **mem_to_reg** (1-bit): Select memory data for writeback.
- **alu_ctrl** (3-bit): ALU operation code (mapped to ALU control inputs).
- **alu_src** (1-bit): Select immediate as ALU second operand.
- **reg_wr** (1-bit): Register file write enable.

NOTE	For load (`LW`), store (`SW`) and branch-equal (`BEQ`) instruction types, the `f3` and `f7` fields do not affect the control signals in this implementation, hence the "don’t care" designation.

4.3.2. Testing Methodology

The test suite provides opcode and function-field patterns and verifies the generated control signals. Test cases exercise:

1. **R-type variants**: Multiple `f3`/`f7` combinations that map to different `alu_ctrl` values and exercise register-write behavior, specifically for the ADD, SUB, AND, and OR instructions in that order.
2. **Load (LW)**: Confirms memory-read path and memory-to-register selection.
3. **Store (SW)**: Confirms memory-write enable and ALU source selection for addressing.
4. **Branch (BEQ)**: Verifies branch output assertion for branch-type opcodes.

Each category uses representative binary encodings from the CSV tests and checks the seven control outputs.

4.3.3. Test Cases and Results

Table 5. R-Type Variants

op	f3	f7	branch	mem_wr	mem_rd	mem_to_reg	alu_ctrl	alu_src	reg_wr
0110011	000	0000000	0	0	0	0	000	0	1
0110011	000	0100000	0	0	0	0	001	0	1
0110011	111	0000000	0	0	0	0	010	0	1
0110011	110	0000000	0	0	0	0	011	0	1

These R-type test cases validate decoding into register-write operations and correct `alu_ctrl` selection for different function fields.

Table 6. Load (LW)

op	f3	f7	branch	mem_wr	mem_rd	mem_to_reg	alu_ctrl	alu_src	reg_wr
0000011	XXX	XXXXXXX	0	0	1	1	000	1	1

This test case confirms that a load instruction asserts `mem_rd`, selects memory data for writeback, and uses the immediate as the ALU source.

Table 7. Store (SW)

op	f3	f7	branch	mem_wr	mem_rd	mem_to_reg	alu_ctrl	alu_src	reg_wr
0100011	XXX	XXXXXXX	0	1	0	0	000	1	0

The store test case verifies `mem_wr` assertion and that the ALU uses the immediate for address calculation; no register write is performed.

Table 8. Branch (BEQ)

op	f3	f7	branch	mem_wr	mem_rd	mem_to_reg	alu_ctrl	alu_src	reg_wr
1100011	XXX	XXXXXXX	1	0	0	0	001	0	0

The branch test case validates that the `branch` control is asserted for a branch-type opcode and that no memory or register writes occur.

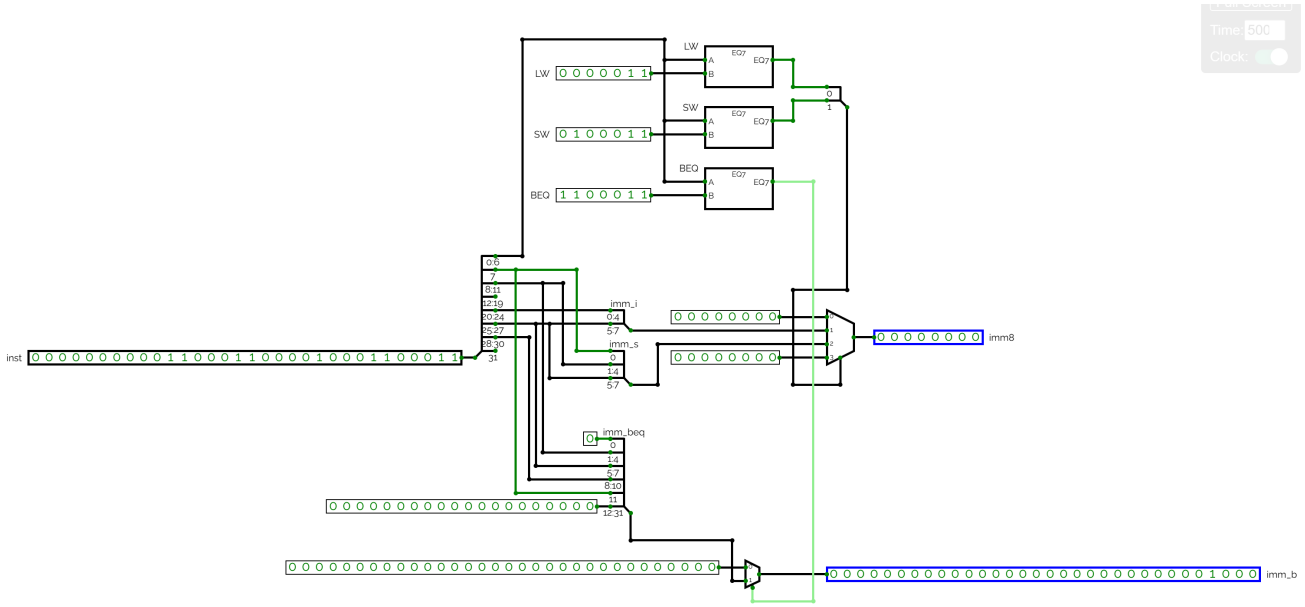
4.3.4. Verification

All test cases produced the expected control outputs. The Control Unit decoding behavior matches the test encodings and the don't-care handling for `f3`/`f7` in LW, SW, and BEQ cases. The component is validated and ready for integration into the processor data path.

4.4. Immediate Generator

The Immediate Generator is a combinational component that extracts immediate values from a 32-bit RISC-V instruction. The generated immediate is used by instructions such as loads, stores, and branches.

4.4.1. Specifications



The Immediate Generator component has the following input and output signals:

Table 9. Immediate Generator Signals

Signal	Bitwidth	Description
`inst`	32	Full 32-bit instruction input.
`imm8`	8	8-bit immediate output used by load and store type instructions.
`imm_b`	32	Branch immediate output used by branch instructions.

The component identifies the instruction type by checking the opcode bits. Based on the instruction format, it selects the correct immediate field:

Table 10. Immediate Format Behavior

Instruction Type	Example	Immediate Behavior
I-type	`lw x1, 0(x0)`	Extracts the immediate from bits `[31:20]` and outputs it through `imm8`.

Instruction Type	Example	Immediate Behavior
S-type	`sw x2, 1(x0)`	Reconstructs the immediate from bits `[31:25]` and `[11:7]`, then outputs it through `imm8`.
B-type	`beq x3, x3, 8`	Reconstructs the branch immediate field and outputs it through `imm_b`.

For the tested branch instruction, the assembly offset is `8` bytes. Internally, this immediate generator outputs the raw branch field as `00000000000000000000000000000100`, which is then used by the datapath branch logic to produce the correct branch target.

4.4.2. Testing Methodology

The Immediate Generator was tested inside CircuitVerse by manually applying known 32-bit instruction encodings to the `inst` input and observing the generated outputs `imm8` and `imm_b`.

The tests were designed to verify the following behavior:

1. Correct immediate extraction for I-type load instructions.
2. Correct immediate reconstruction for S-type store instructions.
3. Correct branch immediate generation for B-type branch instructions.
4. Correct output behavior for instructions that do not use the immediate generator path.

The complete test cases are documented in `tests/Immediate Generator Tests.csv`.

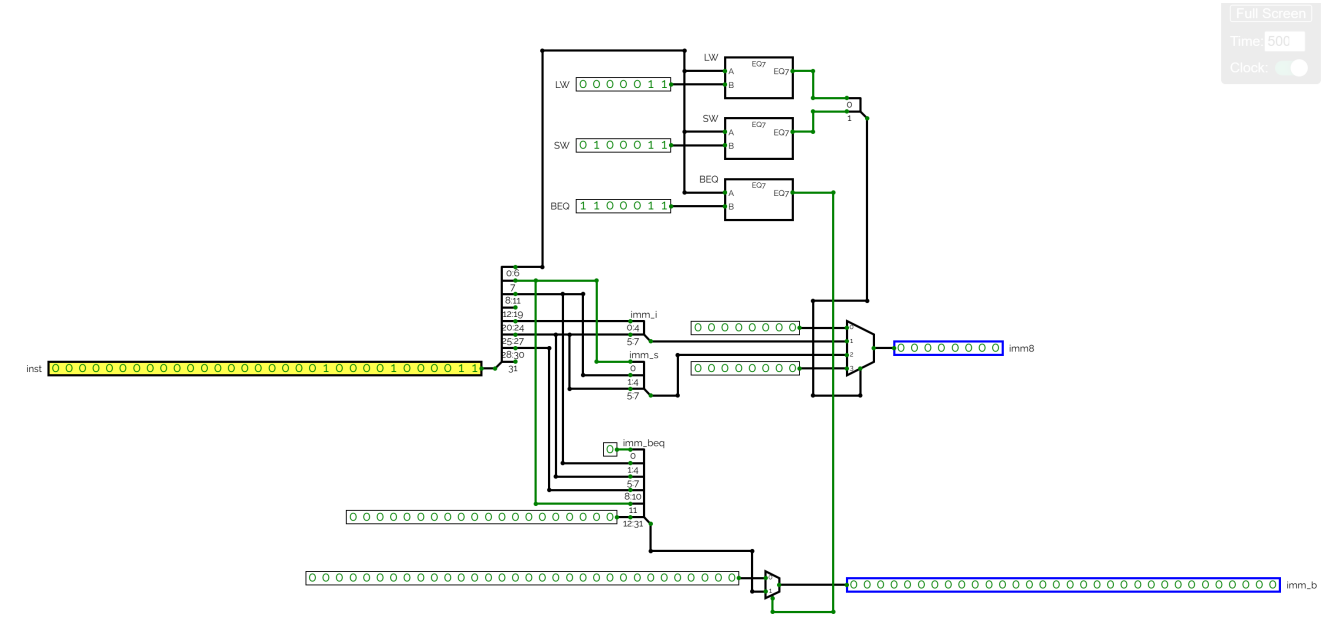
4.4.3. Test Cases and Results

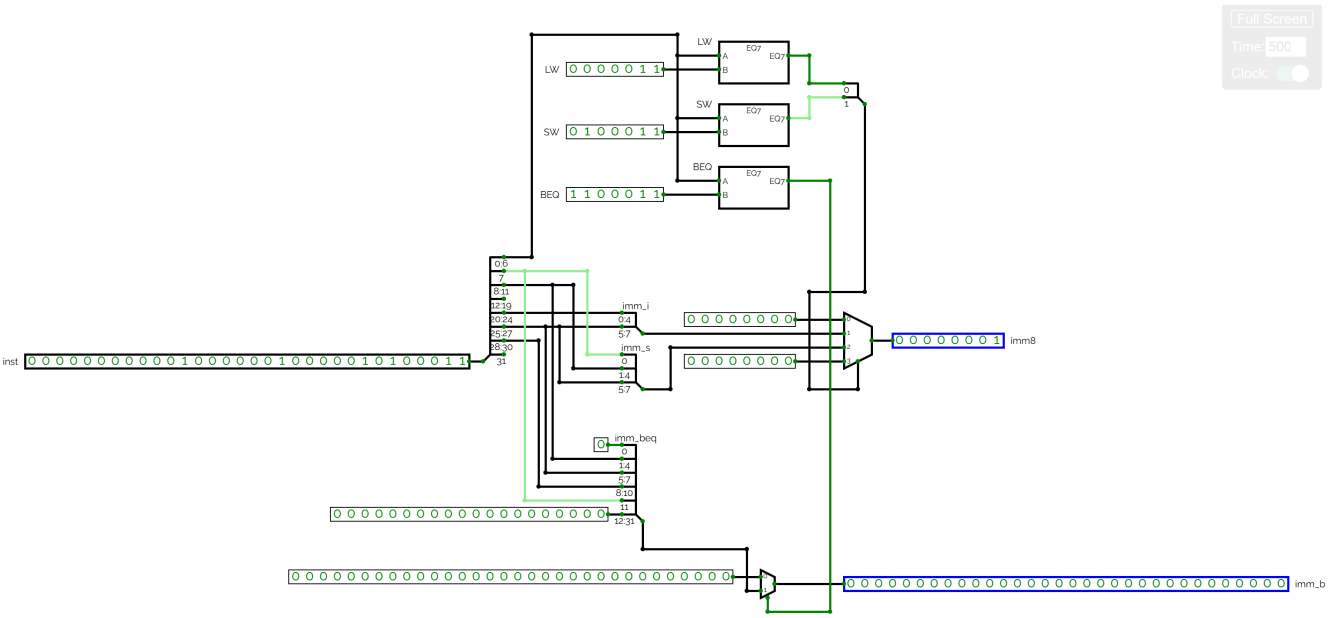
Table 11. Immediate Generator Test Cases

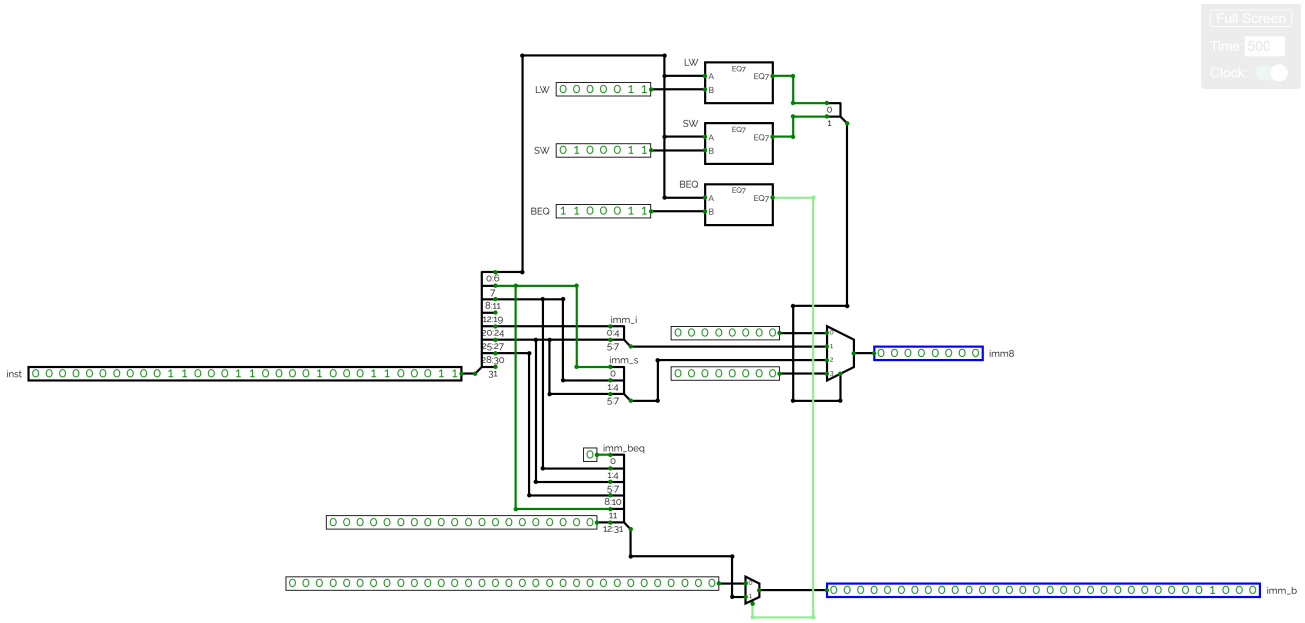
Test	Instruction Input	Expected Output	Result
IG_LW_0	`00000000000000000010000010000011`	`imm8=00000000`, `imm_b=00000000000000000000000000000000`	Passed
IG_SW_1	`00000000001000000010000010100011`	`imm8=00000001`, `imm_b=00000000000000000000000000000000`	Passed
IG_BEQ_8_RAW	`00000000001100011000010001100011`	`imm8=00000000`, `imm_b=00000000000000000000000000000100`	Passed
IG_ADD	`00000000001000001000000100110011`	`imm8=00000000`, `imm_b=00000000000000000000000000000000`	Passed
IG_NOP	`00000000000000000000000000010011`	`imm8=00000000`, `imm_b=00000000000000000000000000000000`	Passed

4.4.4. Evidence

The following images show the internal CircuitVerse tests used to validate the Immediate Generator behavior.







4.4.5. Verification

All test cases produced the expected immediate outputs. The I-type test confirmed that the immediate generator correctly extracts the immediate field for a load instruction. The S-type test confirmed that the component correctly reconstructs the split immediate field used by store instructions.

The B-type test confirmed that the branch immediate field is generated correctly for the `beq` instruction. For `beq x3, x3, 8`, the block outputs the raw branch immediate field as `00000000000000000000000000000100`, matching the observed CircuitVerse behavior. The non-immediate instruction tests confirmed that R-type operations and `nop` do not incorrectly activate the immediate outputs. Therefore, the Immediate Generator component is validated and ready for integration into the processor data path.

4.5. ALU Component

The Arithmetic Logic Unit is a combinational circuit that performs binary arithmetic and logical operations. It serves as the primary computational engine for data manipulation throughout instruction execution.

4.5.1. Specifications

Below we see the ALU circuit in CircuitVerse, with the input and output signals labeled for clarity:

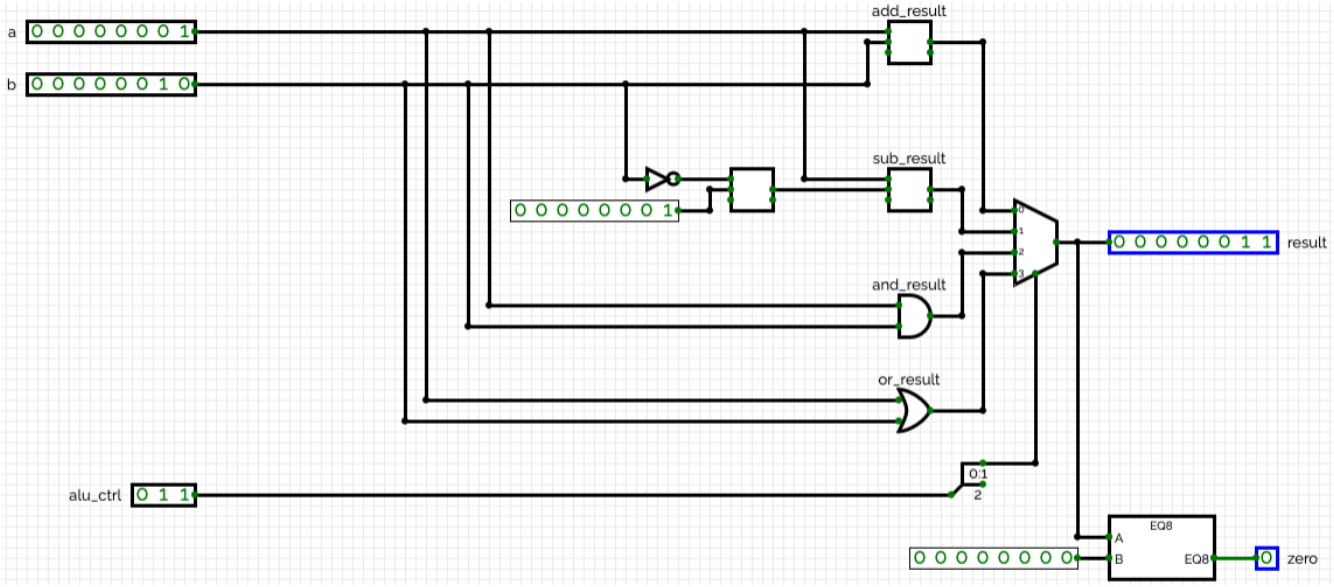


Figure 4. ALU Circuit Diagram

- **a** (8-bit): The first operand for the ALU operation
- **b** (8-bit): The second operand for the ALU operation
- **alu_ctrl** (3-bit): The control signal selecting the ALU operation
- **result** (8-bit): The computed arithmetic or logical result
- **zero** (1-bit): A flag asserted high when the result equals zero

NOTE	The most significant bit of the `alu_ctrl` signal is treated as a "don't care" bit given only 4 operations, representable by 2 bits, are implemented.

The supported operations are defined as follows:

Table 12. ALU Operations

Control Code	Operation	Description
X00	ADD	Binary addition of operands a and b
X01	SUB	Subtraction: a - b using two's complement
X10	AND	Bitwise AND of operands a and b
X11	OR	Bitwise OR of operands a and b

4.5.2. Testing Methodology

The ALU testing protocol exercises all supported operations across diverse operand patterns and edge cases. Six test categories ensure comprehensive coverage:

1. **Addition of Positive Numbers**: Validates binary addition with non-negative operands
2. **Addition of Negative Numbers**: Tests two's complement addition with negative operands
3. **Subtraction of Positive Numbers**: Verifies subtraction including cases producing negative results
4. **Subtraction of Negative Numbers**: Confirms signed subtraction with proper overflow

behavior

5. **Bitwise AND Operation**: Validates bit-by-bit conjunction across operand patterns
6. **Bitwise OR Operation**: Confirms bit-by-bit disjunction including complementary patterns

Each category includes three test cases covering normal operation, zero results, edge values, and patterns exercising all bit positions.

4.5.3. Test Cases and Results

Table 13. Addition of Positive Numbers

Input a	Input b	alu_ctrl	Result	Zero Flag
00000100 (4)	10000010 (130)	X00	10000110 (134)	0
00000101 (5)	00001001 (9)	X00	00001110 (14)	0
00000000 (0)	00000000 (0)	X00	00000000 (0)	1

Addition of positive operands produces correct sums with the zero flag asserted when both inputs are zero.

Table 14. Addition of Negative Numbers

Input a	Input b	alu_ctrl	Result	Zero Flag
11111100 (-4)	11111101 (-3)	X00	11111001 (-7)	0
11111111 (-1)	11111111 (-1)	X00	11111110 (-2)	0
10000000 (-128)	00000001 (1)	X00	10000001 (-127)	0

Two’s complement negative operands are added correctly, with proper overflow handling within the 8-bit domain.

Table 15. Subtraction of Positive Numbers

Input a	Input b	alu_ctrl	Result	Zero Flag
00001010 (10)	00000100 (4)	X01	00000110 (6)	0
00001111 (15)	00001111 (15)	X01	00000000 (0)	1
00000100 (4)	00001010 (10)	X01	11111010 (-6)	0

Subtraction of positive operands generates correct results, including cases where the result is negative.

Table 16. Subtraction of Negative Numbers

Input a	Input b	alu_ctrl	Result	Zero Flag
11111100 (-4)	11111101 (-3)	X01	11111111 (-1)	0
10000000 (-128)	00000001 (1)	X01	01111111 (127)	0
11111111 (-1)	11111111 (-1)	X01	00000000 (0)	1

Signed subtraction using two's complement representation produces correct results including scenarios with sign bit inversion.

Table 17. Bitwise AND Operation

Input a	Input b	alu_ctrl	Result	Zero Flag
11110000	00001111	X10	00000000	1
10101010	01010101	X10	00000000	1
11110011	11001100	X10	11000000	0

Bitwise AND correctly implements conjunction logic. Complementary bit patterns produce zero, while overlapping patterns generate appropriate non-zero results.

Table 18. Bitwise OR Operation

Input a	Input b	alu_ctrl	Result	Zero Flag
11110000	00001111	X11	11111111	0
10101010	01010101	X11	11111111	0
00000001	00000010	X11	00000011	0

Bitwise OR correctly implements disjunction logic. Complementary patterns combine to produce all-ones results, and adjacent single-bit patterns merge correctly.

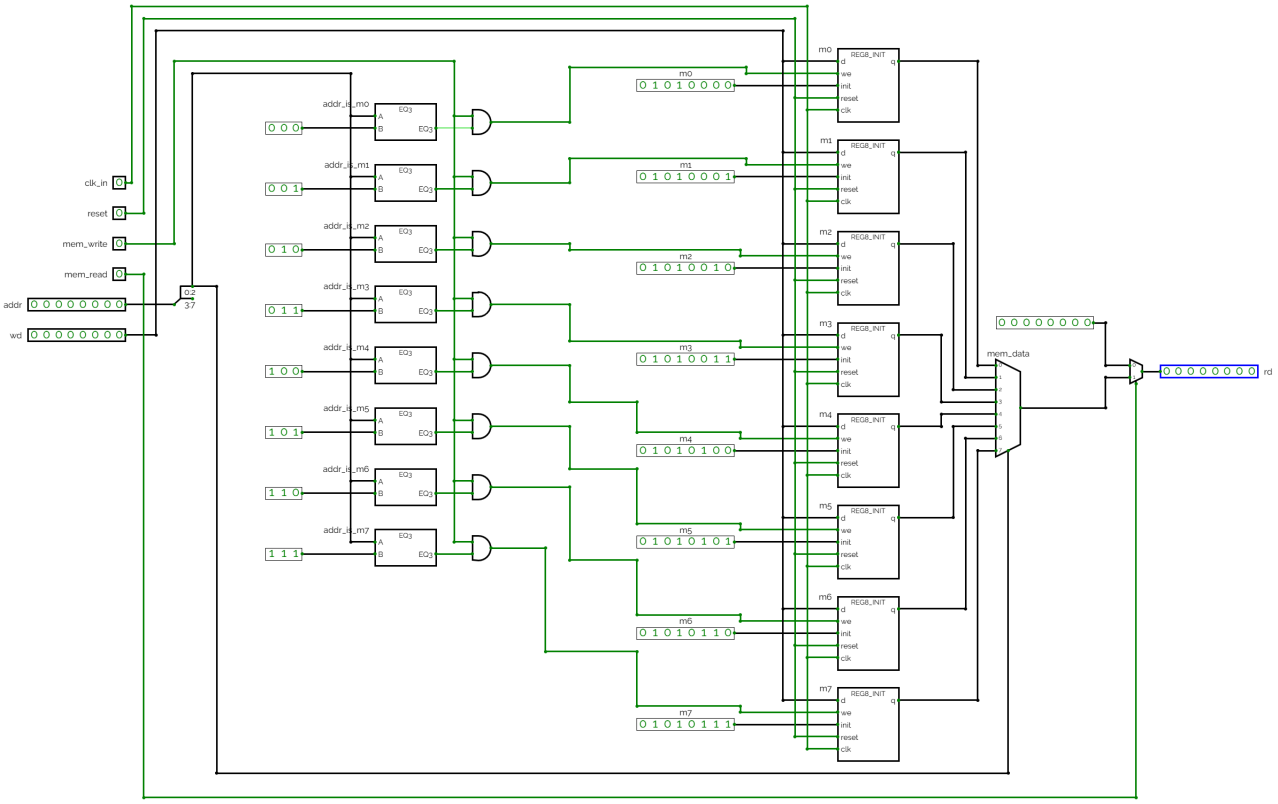
4.5.4. Verification

All ALU test cases execute successfully with results matching predicted values. The ALU correctly implements addition, subtraction, AND, and OR operations with proper two's complement handling and accurate zero flag generation. The component is validated and ready for integration into the processor data path.

4.6. Data Memory

The Data Memory is a sequential component used to store and retrieve data values during load and store instructions. In this simplified processor, the data memory contains eight 8-bit memory locations.

4.6.1. Specifications



The Data Memory component has the following input and output signals:

Table 19. Data Memory Signals

Signal	Bitwidth	Description
`clk_in`	1	Clock input. Memory writes and reset behavior occur synchronously when the clock is triggered.
`reset`	1	Synchronous reset signal. When enabled and the clock is triggered, the memory returns to its initialized values.
`mem_write`	1	Memory write enable signal. When `mem_write = 1`, the value on `wd` is stored at the selected address on a clock pulse.
`mem_read`	1	Memory read enable signal. When `mem_read = 1`, the selected memory value appears at the output `rd`.
`addr`	8	Memory address input used to select the memory location.
`wd`	8	Write data input. This value is written into memory when `mem_write = 1`.
`rd`	8	Read data output.

The memory read output depends on `mem_read`. When `mem_read = 1`, the memory outputs the selected value. When `mem_read = 0`, the output is disabled and returns zero. Writes occur only when `mem_write = 1` and a clock pulse is applied.

4.6.2. Testing Methodology

The Data Memory was tested inside CircuitVerse by manually applying address, write data, read enable, write enable, reset, and clock values.

The tests were designed to verify the following behavior:

1. The memory responds correctly to reset.
2. The memory outputs initialized values when read.
3. A memory location can be written when `mem_write = 1`.
4. The written memory value can be read back correctly.
5. A memory location is not modified when `mem_write = 0`.
6. The read output is disabled when `mem_read = 0`.
7. The memory returns to its initialized values after reset.

The complete test cases are documented in `tests/Data Memory Tests.csv`.

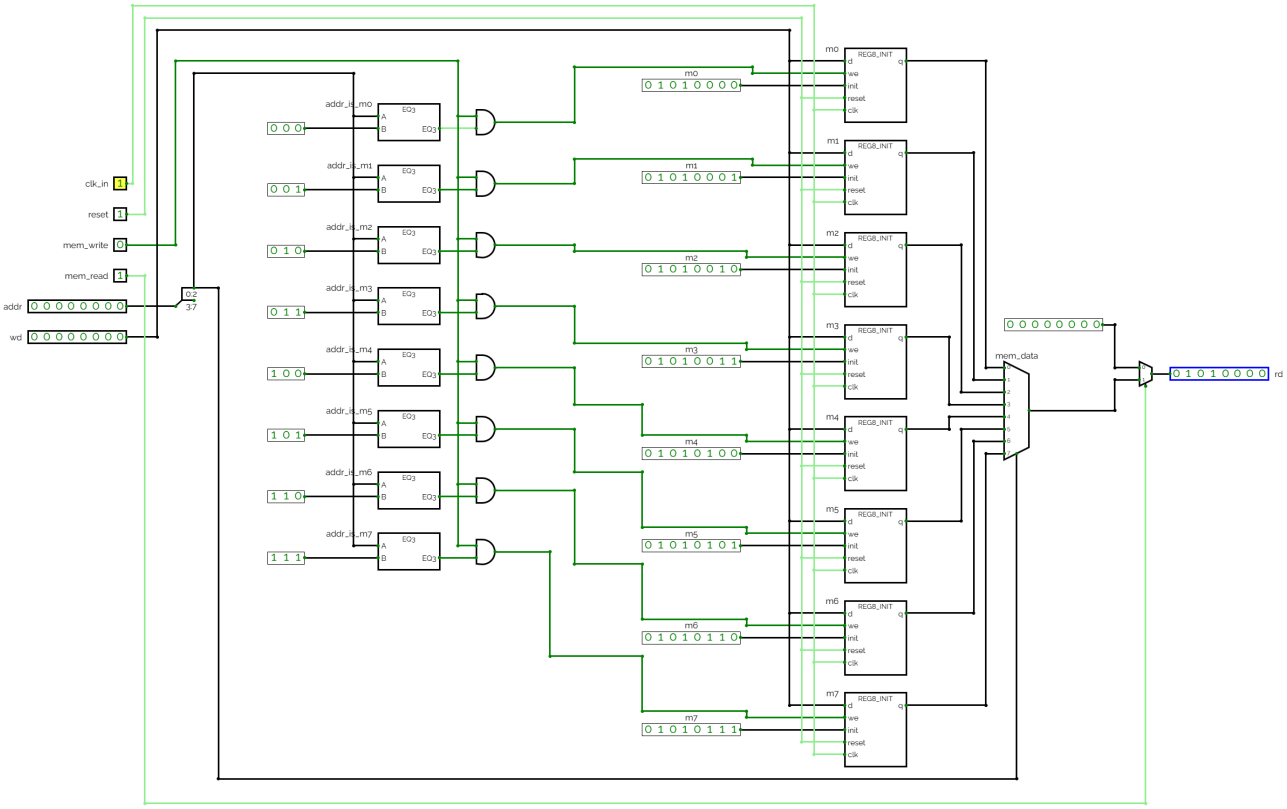
4.6.3. Test Cases and Results

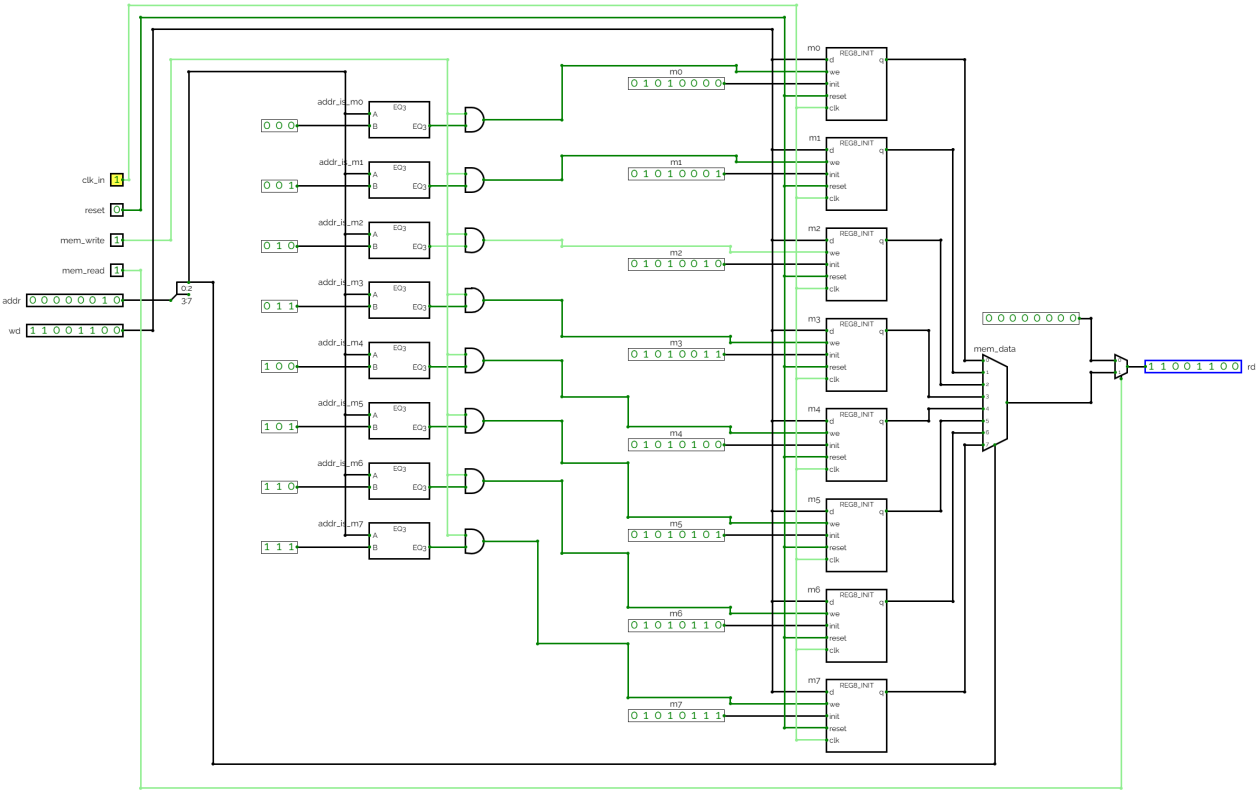
Table 20. Data Memory Test Cases

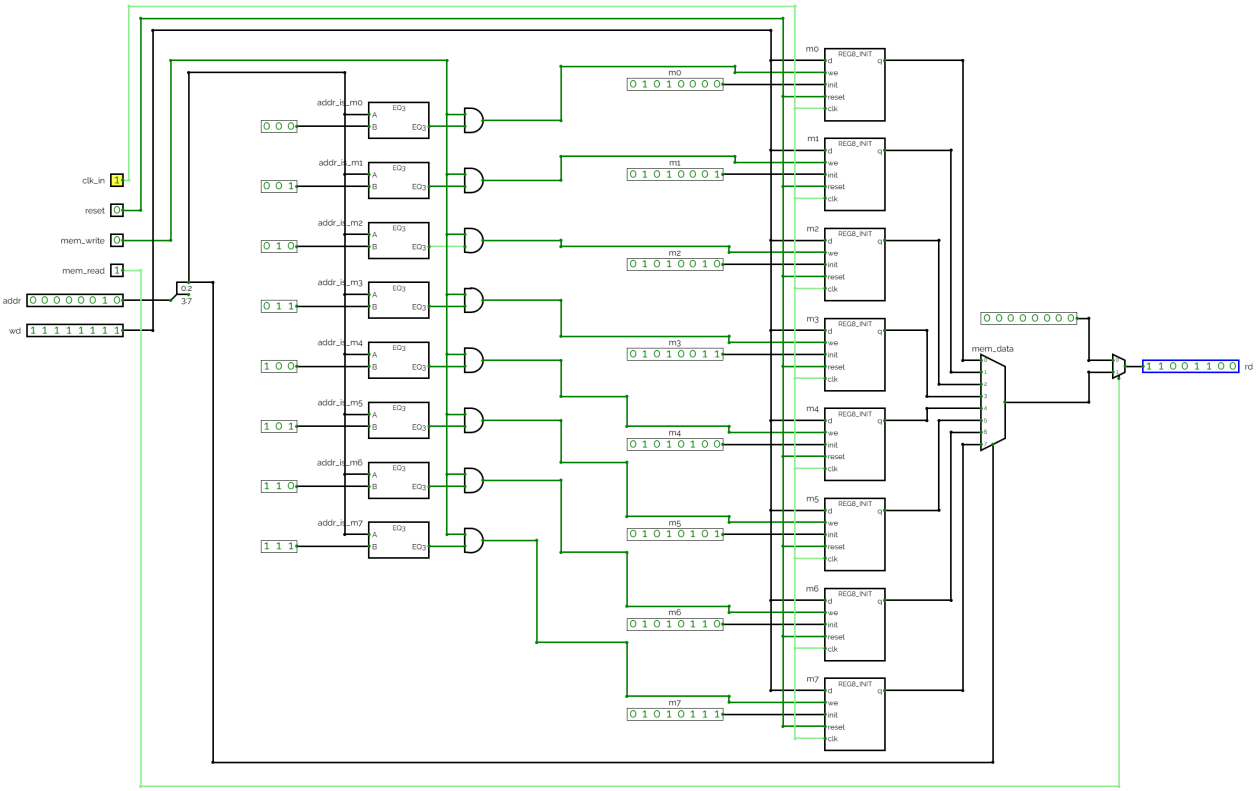
Test	Inputs Applied	Expected Output	Result
DM_RESET_READ_M0	`clk_in=1`, `reset=1`, `mem_write=0`, `mem_read=1`, `addr=00000000`, `wd=00000000`	`rd=01010000`	Passed
DM_READ_M1	`clk_in=0`, `reset=0`, `mem_write=0`, `mem_read=1`, `addr=00000001`, `wd=00000000`	`rd=01010001`	Passed
DM_READ_M7	`clk_in=0`, `reset=0`, `mem_write=0`, `mem_read=1`, `addr=00000111`, `wd=00000000`	`rd=01010111`	Passed
DM_WRITE_M2_CC	`clk_in=1`, `reset=0`, `mem_write=1`, `mem_read=1`, `addr=00000010`, `wd=11001100`	`rd=11001100`	Passed
DM_BLOCK_WRITE_M2	`clk_in=1`, `reset=0`, `mem_write=0`, `mem_read=1`, `addr=00000010`, `wd=11111111`	`rd=11001100`	Passed
DM_READ_DISABLED	`clk_in=0`, `reset=0`, `mem_write=0`, `mem_read=0`, `addr=00000010`, `wd=00000000`	`rd=00000000`	Passed
DM_RESET_AFTER_WRITES_M2	`clk_in=1`, `reset=1`, `mem_write=0`, `mem_read=1`, `addr=00000010`, `wd=00000000`	`rd=01010010`	Passed

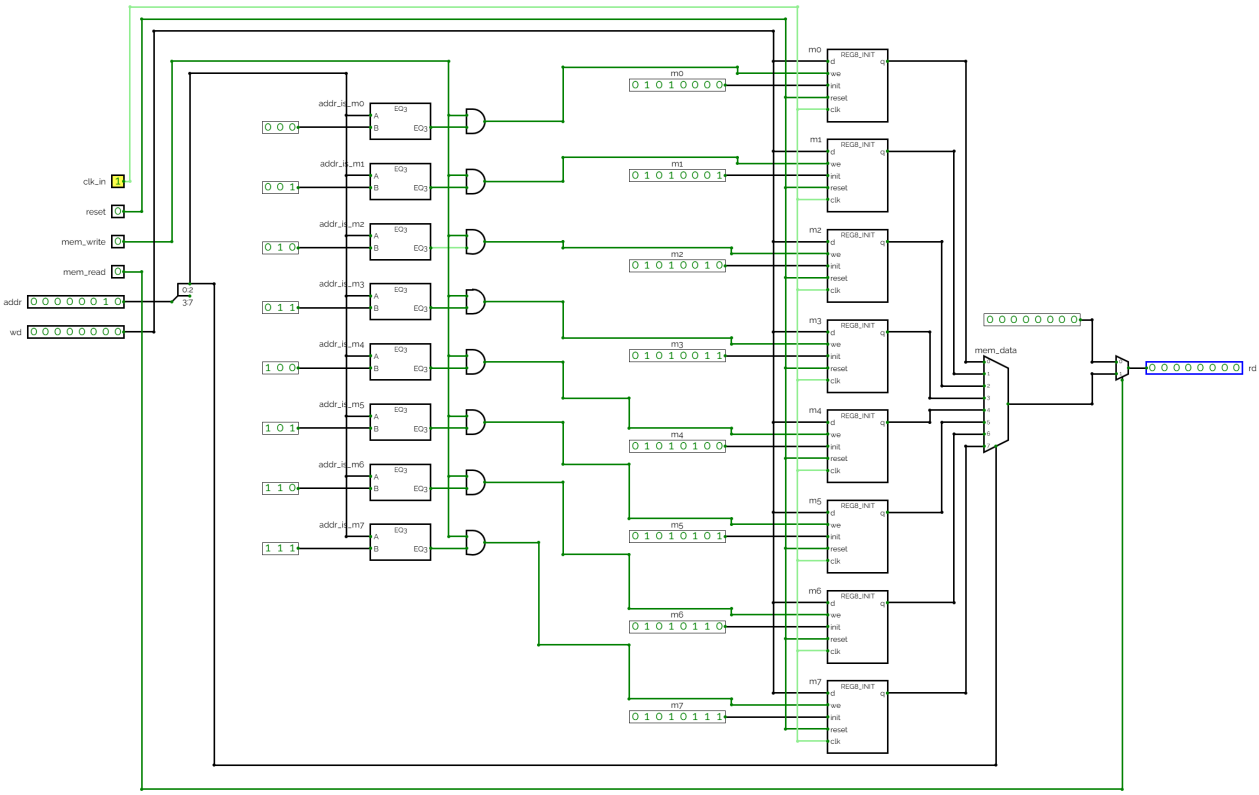
4.6.4. Evidence

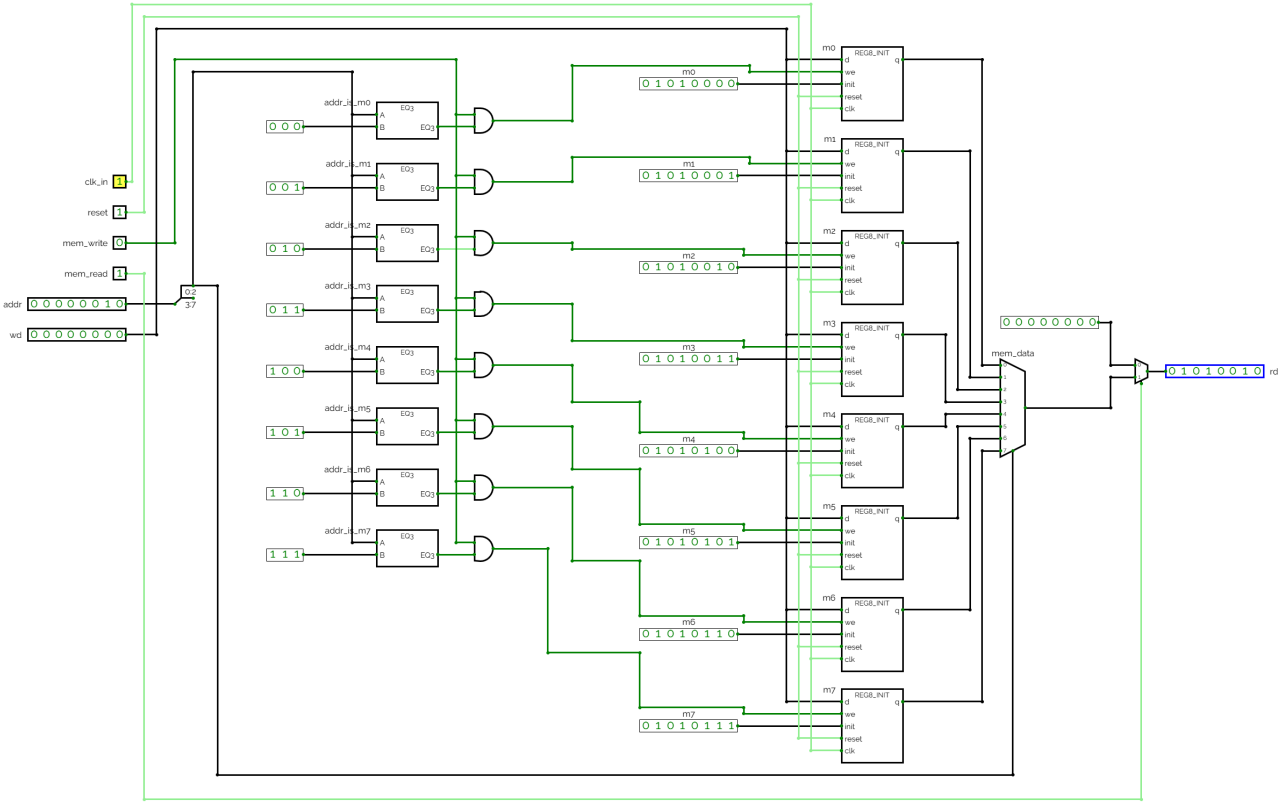
The following images show the internal CircuitVerse tests used to validate the Data Memory behavior.






clk_in
reset
mem_write
mem_read
addr
wd
addr_is_m0
addr_is_m1
addr_is_m2
addr_is_m3
addr_is_m4
addr_is_m5
addr_is_m6
addr_is_m7
m0
m1
m2
m3
m4
m5
m6
m7
REG8_INIT
mem_data
rd


clk_in
reset
mem_write
mem_read
addr
wd
addr_is_m0
addr_is_m1
addr_is_m2
addr_is_m3
addr_is_m4
addr_is_m5
addr_is_m6
addr_is_m7
m0
m1
m2
m3
m4
m5
m6
m7
REG8_INIT
mem_data
rd



4.6.5. Verification

All test cases produced the expected data memory outputs. The reset test confirmed that the memory returns to its initialized values when `reset` is asserted and a clock pulse is applied. The read tests confirmed that the output `rd` correctly reflects the value stored at the selected address when `mem_read = 1`.

The write test verified that memory address `00000010` is updated to `11001100` only when `mem_write = 1` and the clock is triggered. The disabled-write test confirmed that memory contents remain unchanged when `mem_write = 0`, since address `00000010` remained equal to `11001100` even when `wd = 11111111`. The disabled-read test confirmed that the output is cleared to `00000000` when `mem_read = 0`. Finally, the reset-after-writes test confirmed that memory address `00000010` returns to its initialized value `01010010` after reset. Therefore, the Data Memory component is validated and ready for integration into the processor data path.

5. Part 3: Processor Integration and Validation

After verifying individual components, the complete processor must be validated executing the full instruction sequence. This integration phase confirms that components interact correctly and that the processor achieves the intended behavior when executing real programs.

5.1. Testing Protocol

Full processor validation is performed using the CircuitVerse simulation environment shown

below:

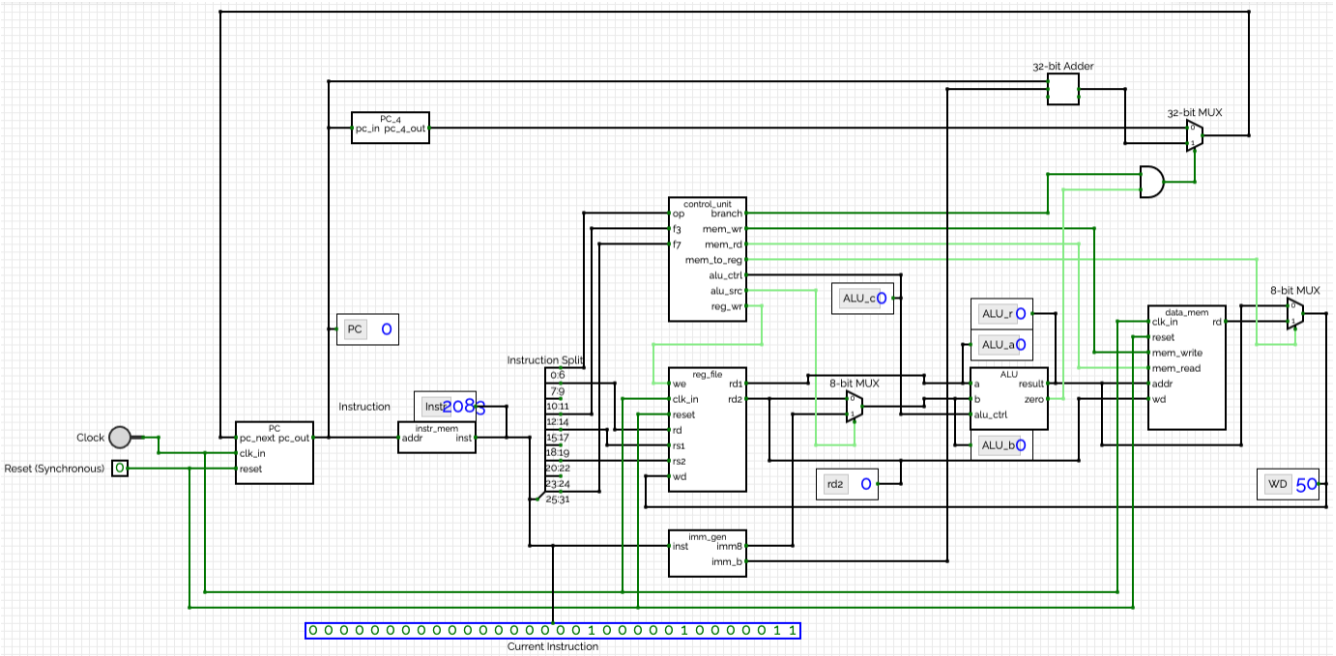


Figure 5. Full Processor Simulation in CircuitVerse

The precise steps are as follows:

1. Set the reset signal to (1), and force a single clock cycle to initialize the processor state.
2. Set the reset signal back to (0) to allow normal operation.
3. Force clock cycles one by one and observe the processor executing the instruction sequence defined in the instruction memory from the timing diagram view, monitoring key signals and outputs at each clock cycle.
4. Compare the observed behavior against expected results based on the instruction sequence and initial state of the processor.

For context, the contents of the instruction memory, register file, and data memory at the start of execution are as follows:

Table 21. Instruction Memory

Address	Instruction (Hex)	Instruction (Assembly)
0x00	0x00002083	lw x1, 0(x0)
0x04	0x00208133	add x2, x1, x2
0x08	0x002020A3	sw x2, 1(x0)
0x0C	0x00318463	beq x3, x3, 8
0x10	0x40638233	sub x4, x7, x6
0x14	0x0020E2B3	or x5, x1, x2
0x18	0x00000013	addi x0, x0, 0(nop)
0x1C	0x00000013	addi x0, x0, 0(nop)

Table 22. Register File

Register	Initial Value (Hex)
x1	0x00000029
x2	0x0000002A
x3	0x0000002B
x4	0x0000002C
x5	0x0000002D
x6	0x0000002E
x7	0x0000002F

Table 23. Data Memory

Address	Initial Value (Hex)
0x00	0x00000050
0x01	0x00000051
0x02	0x00000052
0x03	0x00000053
0x04	0x00000054
0x05	0x00000055
0x06	0x00000056
0x07	0x00000057

NOTE Hexadecimal values are used for clarity, because the CircuitVerse simulation displays values in hex when using the timing diagram and its corresponding "Flag" elements.

5.2. Signal Monitoring Points

The following signals are sufficient to verify correct execution of the given instruction sequence:

- Program Counter (**PC**): Current instruction address
- Current Instruction (**Instr**): Instruction fetched from instruction memory
- Register x1 (**x1**): Destination of the `lw` instruction
- Register x2 (**x2**): Destination of the `add` instruction
- Data Memory Address 0x01 (**mem[0x01]**): Destination of the `sw` instruction
- Register x4 (**x4**): Destination of the skipped `sub` instruction
- Register x5 (**x5**): Destination of the `or` instruction executed after the branch

Monitoring these values allows verification of instruction fetch, arithmetic execution, memory access, branch behavior, the absence of writes from the skipped `sub` instruction, and correct execution of the subsequent `or` instruction.

5.3. Instruction Execution Verification

The expected state transitions for the monitored values are shown below.

Table 24. Expected Results

PC	Instr	x1	x2	mem[0x01]	x4	x5
0x00	lw x1, 0(x0)	0x00000050	0x0000002A	0x00000051	0x0000002C	0x0000002D
0x04	add x2, x1, x2	0x00000050	0x0000007A	0x00000051	0x0000002C	0x0000002D
0x08	sw x2, 1(x0)	0x00000050	0x0000007A	0x0000007A	0x0000002C	0x0000002D
0x0C	beq x3, x3, 8	0x00000050	0x0000007A	0x0000007A	0x0000002C	0x0000002D
0x14	or x5, x1, x2	0x00000050	0x0000007A	0x0000007A	0x0000002C	0x0000007A
0x18	nop	0x00000050	0x0000007A	0x0000007A	0x0000002C	0x0000007A
0x1C	nop	0x00000050	0x0000007A	0x0000007A	0x0000002C	0x0000007A

5.4. Processor Validation Results

After executing the instruction sequence and monitoring the key signals, the following timing diagram is obtained, showing the observed values for the monitored signals at each clock cycle:

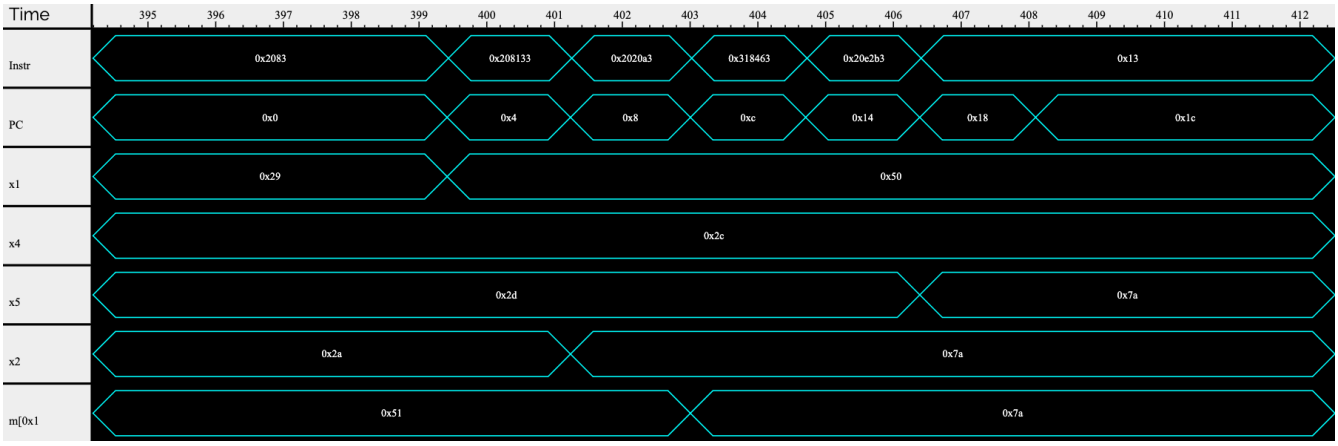


Figure 6. Timing Diagram of Full Processor Execution

NOTE The data is written at the end of the clock cycle, so the observed values for x1, x2, mem[0x01], x4, and x5 reflect the updated state after the instruction execution for that cycle.

The observed values obtained from the timing diagram above are compared against the expected values below:

Table 25. Validation Summary

Verification Item	Expected Result	Observed Result	Status
Final value of x1	0x00000050	0x00000050	Pass
Final value of x2	0x0000007A	0x0000007A	Pass

Verification Item	Expected Result	Observed Result	Status
Final value of mem[0x01]	0x0000007A	0x0000007A	Pass
Final value of x4	0x0000002C	0x0000002C	Pass
Final value of x5	0x0000007A	0x0000007A	Pass
PC value after 0x0C (beq)	0x14	0x14	Pass

Agreement between the observed values and the expected values confirms correct execution of the instruction sequence and successful integration of all processor components.

6. Conclusion

Component-level testing confirmed the correct operation of individual subsystems, while integration testing verified proper interaction within the complete processor pipeline.

The structured approach to validation, testing components in isolation before system-wide integration, ensures that the processor architecture is sound and each element functions as designed. The comprehensive test protocols provide clear evidence of functional correctness and serve as a foundation for confident deployment of the processor design in subsequent applications.

All major processor subsystems have been verified to meet their functional specifications. The documented test results provide a complete record of validation activities and demonstrate the processor's readiness for operation with the full instruction set.

Once again, the entire evolution of this project, including all testing and validation activities, is available in our GitHub repository for reference and further exploration.